Assignment 1

Student Project Approval & Allocation System

Complete BPMN model, happy path, failure register and notation rationale. Only Assignment 1 is included.

Scope and responsibilities

The case begins when a student team decides to submit a final-year project proposal. It ends successfully only after formal guide acceptance, an authoritative allocation record, archived evidence and delivered student notification. Unsuccessful outcomes are rejected, lapsed or withdrawn, each with a reason and audit trail. All five participants are represented as lanes in one departmental process: Student / Team, Project Coordinator, Review Committee / HoD, Faculty Guide and Project Management System.

Happy path

1. Team submits proposal and abstract.
2. System checks fields, documents, identity and team rules.
3. Coordinator screens feasibility and eligibility.
4. System checks topic similarity; coordinator adjudicates suspicious matches.
5. Committee evaluates and approves the proposal.
6. Business rules rank suitable guides by domain, workload and conflicts.
7. System atomically reserves a guide slot and sends the allocation request.
8. Guide formally accepts.
9. System commits the confirmed allocation; notification and evidence archiving run in parallel.
10. Both branches complete, and the case ends as allocated and notified.

How to read the diagrams

Page 1 shows overall completion, closure, withdrawal and escalation. Page 2 shows the six stages. Pages 3-8 detail each stage. Validation and screening are reusable called processes: changed and new proposals invoke these same complete checks, preserving case counters. The editable .bpmn file contains all eight BPMN diagram planes. Open it in a BPMN 2.0 editor and navigate subprocesses / called processes, or choose a diagram where supported. SVG and PNG files provide independent previews. Sequence flows cross responsibility lanes; message events represent withdrawal requests. No message flows are drawn between lanes within one process.

Proposed rules and assumptions

The brief does not set numerical policies. All deadlines and limits below are proposed design values, not claimed university rules. Replace them with department policy before implementation.

Submission opens for 14 days from case creation; coordinator may grant one 3-day extension. Validation corrections and topic changes each have a maximum of two cycles, with 7 days per response.

Committee revisions: at most two cycles. A final approve/reject decision follows within 5 days. One new topic after rejection is allowed within 7 days; case counters are retained, preventing an unlimited restart.

Review SLA per review activation: reminder at day 3, HoD escalation at day 5, hard lapse at day 10. Quorum and conflict checks are part of the committee task. Replacement reviewers preserve the same deadline.

Guide response: 3 days. After three failed candidate attempts, use manual matching. Manual matching / waitlist lasts at most 14 days. The guide roster is finite; excluded guides are not retried in the same case. External or co-guides still need formal acceptance and policy approval.

Technical validation and final database / notification operations allow three total automated attempts, counting the initial attempt, with 5-minute intervals. Persistent failure uses a controlled signed manual register or verified human contact. Technical failures in other stages go directly to the documented human fallback.

Manual allocation records are authoritative when the system is down and must be reconciled later under the same case key. A notification is complete only with delivery evidence; unread email alone is not treated as proof. Unreachable students lapse after 7 days.

Team withdrawal is accepted throughout the active approval case. Guide withdrawal is caught during allocation and finalisation; cleanup voids provisional confirmation and returns the case to allocation. Guide changes after successful case closure are a separate project-management process.

Task types and gateways

User tasks

Proposal entry, screening, committee decisions, guide acceptance and manual fallbacks require a person to act through an interface or controlled register. Changed proposals subsequently invoke the reusable validation and screening processes, with the correct system and coordinator responsibilities.

Service tasks

Validation, similarity lookup, atomic reservation, durable recording, cleanup and archiving are automated operations. Technical errors have explicit recovery boundaries where identified.

Business rule task

Guide ranking applies documented domain, workload and conflict rules. It does not reserve capacity; a later atomic service task avoids races.

Script task

Initialise the case counters and audit context at case opening. Later proposal changes keep those same counters; they never initialise a fresh case budget.

Send tasks

Allocation requests and student notifications create a durable outbound record and log delivery. A user task captures the guide decision because it requires an actual human decision, not just arrival of a message.

Exclusive gateways

Validation outcomes, extension approval, suitability, similarity adjudication, committee decisions, loop budgets, candidate availability and acceptance are mutually exclusive. Each split has a default route and labelled conditions; default routes resolve unexpected or unsuccessful outcomes conservatively.

Parallel gateways

After recording, student notification and evidence archiving must both happen. The split creates two branches, and the join waits for both. Alternative manual branches merge into those same completion inputs; only one route per branch arrives.

Why no inclusive or event-based gateway

No stage needs an arbitrary subset of optional concurrent checks. Guide acceptance is a user decision with an interrupting timer boundary, so an event-based race would duplicate that construct. The rubric asks for correct gateway use; unnecessary gateway types would make this model harder to justify.

Subprocesses, calls and errors

Collapsed subprocesses keep the main path readable. Thick-bordered call activities invoke the reusable validation and screening processes. Case data and counters are shared by reference as a modelling contract. CASE_CLOSED error end events propagate through called activities, abort the approval scope including parallel work, and activate its error boundary. The outer handler records the reason and delivers closure information. Technical error boundaries interrupt the failed operation.

Timers and escalation

Submission, correction, revision, guide-response and fallback deadlines interrupt their activity. Review reminders and escalation timers are non-interrupting so the review remains active. The escalation throw is caught on the enclosing approval scope, where the HoD expedites the review.

Compensation and withdrawal

A compensation boundary on the completed Reserve task is associated with an idempotent Release task marked isForCompensation. Guide decline / silence throws targeted compensation before excluding that candidate. Cross-scope team or guide withdrawal uses explicit idempotent cleanup: an interrupted child scope is not assumed to retain its internal compensation handlers.

Failure Path Register

Each row maps a failure to exact editable BPMN IDs and its resolution. See the CSV for sorting and filtering.

F01 | Submission - No proposal before deadline

Detect: Submit_Deadline
Resolve: One late request; 3-day extension or lapse
Owner: Coordinator
Limit: 14-day initial window; one extension
Outcome: Submit_Lapsed

F02 | Submission - Late request denied / extended deadline missed

Detect: Submit_LateOK; Submit_Expired
Resolve: Record lapse and deliver reasons through outer closure handler
Owner: Coordinator / System
Limit: 3-day extension
Outcome: Main_EndClosed

F03 | Validation - Missing abstract, unreadable file or incomplete form

Detect: Validate_Result; Validate_Limit
Resolve: Return error list; two corrections; coordinator resolves or rejects
Owner: Student / Coordinator
Limit: 7 days per correction; max 2 cycles
Outcome: Validate_End / Validate_Invalid

F04 | Validation - Invalid roll numbers, team size or duplicate team membership

Detect: Validate_Result
Resolve: Reject invalid team with reasons; team may reform in a separate case
Owner: System / Coordinator
Limit: No invalid team progresses
Outcome: Validate_Invalid

F05 | Validation - Correction not submitted

Detect: Validate_CorrectionDue
Resolve: Close as lapsed; notify with recorded reason
Owner: System
Limit: 7 days
Outcome: Main_EndClosed

F06 | Validation - Validation service / upload store unavailable

Detect: Validate_Error; Validate_Retry
Resolve: Max three automated attempts at five-minute intervals; signed manual validation
Owner: System / Coordinator
Limit: Initial attempt included in total 3
Outcome: Validate_Manual

F07 | Screening - Infeasible scope, missing resources or ineligible topic

Detect: Screen_Suitable; Screen_Adjudicate
Resolve: Review reasons; at most two topic changes with revalidation; otherwise reject
Owner: Coordinator / Student
Limit: 7 days; max 2 changes
Outcome: Screen_Rejected

F08 | Similarity - Duplicate or suspected plagiarised topic

Detect: Screen_Match; Screen_CanRevise
Resolve: Human evidence review; justified topic change or reject; no automatic plagiarism conviction
Owner: Coordinator
Limit: Same topic-change budget
Outcome: Screen_Revise / Screen_Rejected

F09 | Similarity - Similarity API unavailable

Detect: Screen_Tech
Resolve: Manual repository comparison with evidence, then same decision gateway
Owner: Coordinator
Limit: One documented fallback
Outcome: Screen_Manual

F10 | Screening - Topic revision not submitted

Detect: Screen_Due
Resolve: Lapse after bounded revision period
Owner: Student / System
Limit: 7 days
Outcome: Screen_Rejected

F11 | Review - Committee requests revision

Detect: Review_RevLimit
Resolve: Two revision cycles; then mandatory approve/reject decision
Owner: Committee / Student
Limit: 7 days each; final decision 5 days
Outcome: Review_FinalDecision

F12 | Review - Revised proposal not submitted

Detect: Review_RevDue
Resolve: Close as lapsed, communicate reasons
Owner: System
Limit: 7 days
Outcome: Review_Closed

F13 | Review - Committee rejects proposal

Detect: Review_NewAllowed
Resolve: Allow one fully checked new topic; further rejection closes case
Owner: Committee / Student
Limit: New topic once, within 7 days; counters retained
Outcome: Review_NewTopic / Review_Closed

F14 | Review - Review delayed, no quorum or conflicted reviewers

Detect: Review_Reminder; Review_Escalate; Review_HardDue
Resolve: Reminder at day 3; escalation at day 5; HoD reassigns; hard closure at day 10
Owner: Coordinator / HoD
Limit: Timers run relative to each review task activation
Outcome: Main_Expedite / Review_Closed

F15 | Review - No final decision / no new-topic submission

Detect: Review_FinalDue; Review_NewDue
Resolve: Close as lapsed; no unlimited new-topic or decision loop
Owner: Committee / Student
Limit: 5 / 7 days
Outcome: Review_Closed

F16 | Allocation - No matching guide or all at maximum workload

Detect: Allocate_Available; Allocate_Manual
Resolve: Coordinator chooses qualified co-guide / external guide or waits; lapse if unresolved
Owner: Coordinator
Limit: 14-day manual/waitlist maximum
Outcome: Allocate_Selected / Allocate_Lapsed

F17 | Allocation - Guide declines or does not respond

Detect: Allocate_Accepted; Allocate_NoReply
Resolve: Compensate reserved slot, exclude guide and try next; manual allocation after 3 failed attempts
Owner: Guide / System / Coordinator
Limit: 3-day response; 3 automated attempts
Outcome: Allocate_Undo; Allocate_Exclude

F18 | Allocation - Guide rule service unavailable

Detect: Allocate_RuleError
Resolve: Manual matching with same domain, load and conflict checks
Owner: Coordinator
Limit: 14-day limit on manual matching
Outcome: Allocate_Manual

F19 | Allocation - Concurrent allocation takes last slot / uncertain DB commit

Detect: Allocate_Conflict
Resolve: Reconcile case-key hold; release if present; increment attempts; rerun capacity checks
Owner: Coordinator / System
Limit: Atomic reservation; idempotent case key
Outcome: Allocate_Reconcile

F20 | Allocation - Allocation request notification fails

Detect: Allocate_SendError; Allocate_ManualAcceptDue
Resolve: Coordinator contacts guide and records request; absence after 3 days compensates slot
Owner: Coordinator
Limit: 3-day manual contact deadline
Outcome: Allocate_Contact / Allocate_Undo

F21 | Allocation - Guide withdraws before process completion

Detect: Case_GuideWithdraw; Case_GuideWithdrawFinal
Resolve: Interrupt allocation or finalisation, release hold, void confirmation, exclude guide and reallocate
Owner: System / Coordinator
Limit: Active allocation and finalisation scopes
Outcome: Case_Release

F22 | Whole case - Student team withdraws or dissolves

Detect: Main_Withdraw
Resolve: Interrupt active case; release any hold, void pending records, audit and confirm withdrawal
Owner: System / Coordinator
Limit: Available throughout active approval case
Outcome: Main_EndWithdraw

F23 | Finalisation - Confirmed allocation cannot be written to database

Detect: Finalize_DBError; Finalize_Retry
Resolve: Three total automated attempts; controlled signed manual register counts as authoritative fallback
Owner: Coordinator / System
Limit: 5-minute intervals; initial attempt included
Outcome: Finalize_ManualRecord

F24 | Finalisation - Email / student notification delivery fails

Detect: Finalize_NotifyError; Finalize_NoticeRetry
Resolve: Three total attempts; verified phone or portal fallback; delivery evidence required
Owner: Coordinator / System
Limit: 5-minute retry delay; then 7-day manual limit
Outcome: Finalize_ManualNotify

F25 | Finalisation - Archiving service fails

Detect: Finalize_ArchiveError
Resolve: Store signed evidence in controlled manual register; join only after archive completed
Owner: Coordinator
Limit: Required before successful completion
Outcome: Finalize_ManualArchive

F26 | Finalisation - Student unreachable after fallback

Detect: Finalize_ContactDue
Resolve: Release guide hold, void confirmation, close as lapsed; no false success
Owner: Coordinator / System
Limit: 7 days
Outcome: Finalize_Release; Finalize_Lapsed

Verification and implementation notes

Static verification

The XML is validated against the official OMG BPMN 2.0 XSD and diagram schemas. Checks also verify unique IDs, resolvable flow / boundary / event references, lane membership and the parallel gateway branch structure. Every register detection ID is present in the model. Diagram PDFs are rendered for visual review. This is a descriptive model (isExecutable=false), not a deployed workflow application.

Example walkthroughs

Happy path: Main_Start -> Main_Run -> Case_Submit -> Case_Validate -> Case_Screen -> Case_Review -> Case_Allocate -> Case_Finalize -> Main_Success.
Invalid team: Validate_Result -> Validate_Invalid error end -> Main_Closed -> Main_Close -> Main_Notice -> Main_EndClosed.
Guide silence: Allocate_NoReply -> Allocate_Undo -> associated Allocate_Release -> Allocate_Exclude -> Allocate_Rule.
Review delay: Review_Escalate -> Review_ThrowEsc; Main_Esc -> Main_Expedite, while Review_Review remains active.
Email failure: Finalize_NotifyError -> bounded notice retry -> Finalize_ManualNotify -> Finalize_Join after archiving also finishes.
Team withdrawal: Main_Withdraw interrupts approval -> Main_Undo -> Main_WithdrawAudit -> Main_EndWithdraw.

Data and operational contract

Case state includes team ID, proposal version, validation issues, decision reason, correction count, topic-change count, revision count, new-topic-used count, guide-attempt count, excluded-guide IDs, operation-specific technical attempt counts, reserved-slot ID, acceptance evidence and delivery proof. Increment each counter on its labelled failure or revision; initialise technical counters per operation activation. Persist counts and audit entries atomically. Late or duplicate callbacks must be ignored for closed cases using case status and an idempotency key. Closure reasons must be set before throwing CASE_CLOSED. A signed manual record must be reconciled into the system without creating a second allocation.

Practical limits

The model covers the 26 registered failure categories, not every imaginable infrastructure fault. Manual tasks require department escalation procedures; their completion cannot be assumed while their systems or responsible staff remain unavailable. Automatic compensation is shown for successful reservations inside allocation; other scope interruptions use explicit cleanup. System-wide disaster recovery and post-completion guide replacement fall outside the assignment boundary.

Sources

Assignment scenario and evaluation criteria: supplied BPMN_Project_Assignments.pdf, Assignment 1 and evaluation rubric. The instructor reference pages are not redistributed in this bundle. Notation and XML schemas: Object Management Group, Business Process Model and Notation 2.0, https://www.omg.org/spec/BPMN/2.0/.